

CSE208: Problem on Intractability

ID-1:

ID-2:

Problem Definition

- **Dominating Set (DS):** In a graph $G = (V, E)$, a subset of vertices $D \subseteq V$ is a dominating set if every vertex $v \in V \setminus D$ is adjacent to at least one vertex in D .
- **Connected Dominating Set (CDS):** A subset of vertices $D \subseteq V$ is a connected dominating set if it is a dominating set, and the subgraph induced by D (the graph containing only the vertices of D and the edges between them) is connected.
- **CDS Decision Problem:** Given a graph G and an integer k , does there exist a connected dominating set of size at most k ?

Task 1: Give an example of a dominating set that is not connected. Does there exist any graph that has no connected dominating set?

First, we need to prove that CDS decision problem is in NP. For a given instance, we need to check if that set is both dominating and connected in polynomial time.

Task 2: Prove that CDS decision problem is in NP.

Now, we will show that CDS decision problem is NP-hard. We will choose a renowned NP-complete problem (Vertex Cover problem), reduce an arbitrary instance of Vertex Cover problem to an "equivalent" instance of CDS decision problem. By 'equivalent', we mean that the original instance of Vertex Cover problem is satisfiable if and only if the transformed instance of CDS Decision Problem is satisfiable

Task 3: Describe how to transform an instance of Vertex Cover problem into an 'equivalent' instance of CDS Decision Problem. You also have to show this transformation runs in polynomial time.

Task 4: Prove that the transformation you described in Task 3 is indeed a polynomial-time reduction from Vertex Cover to CDS Decision Problem. In other words, show that,

- If the original Vertex Cover instance is satisfiable, then the transformed CDS instance is also satisfiable.
- If the original Vertex Cover instance is not satisfiable, then the transformed CDS instance is also not satisfiable.

OR equivalently, If the transformed CDS instance is satisfiable, then the original Vertex Cover instance is also satisfiable.